

MINOR PROJECT 3

Project Title : Predicting Heart Disease (Classification)

1.Introduction :

Brief description of the project and the dataset used.

Heart disease is one of the major causes of death worldwide. Early prediction of heart disease can help doctors take preventive action and provide better treatment. In this project, a Machine Learning model was developed to predict whether a patient has heart disease or not using the Logistic Regression algorithm.

The dataset used for this project is the **UCI Heart Disease Dataset**, a well-known healthcare dataset used for classification problems. It contains medical features such as age, sex, chest pain type, blood pressure, cholesterol, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, and other clinical attributes. The target column indicates whether the patient has heart disease or not.

The project was implemented using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. Data preprocessing steps such as handling missing values, train-test split, feature scaling using StandardScaler, and model training were performed.

2. Exploratory Data Analysis

Before building the model, I explored the data to find patterns and fix problems. This is called EDA.

- **Cleaning the Messy Data:** I found that many medical records were missing information (like cholesterol levels).

- I handled the missing values by filling numerical columns with their median values and categorical columns with their mode values. This helped maintain a complete dataset for further analysis and model training.
- Checking for Balance: I made a chart to see if I had enough examples of both healthy and sick people.
- The data was well-balanced, which means the model could learn fairly from both groups.
- Finding the Best Clues: I used a Correlation Heatmap (a color-coded chart) to see which medical factors were most linked to heart disease.
- This showed me that things like Chest Pain and Heart Stress (Oldpeak) were the most important "red flags" to watch out for.
- Age Analysis: I created a graph that compared age to heart disease.
- It clearly showed that as patients got older (especially after 50), the **number of heart disease cases went up significantly.**

3. Visualization

Visualization (The "Why" Behind the Data)

The visualization phase was used to validate medical assumptions and identify the most predictive features for the model.

- **Class Balance (Countplot):** We first checked the distribution of the target variable. The dataset was well-balanced (approx. 55% Heart Disease vs. 45% Healthy), which ensured that the model wouldn't be biased toward a single outcome.

- **Correlation Heatmap:** This was the most critical visual tool. By plotting a heatmap of numerical factors, we identified that Oldpeak (ST depression) and Max Heart Rate (thalch) had the strongest statistical relationships with heart disease. This allowed us to prioritize these features for the Logistic Regression model.
- **Age Distribution (Histogram):** By overlaying histograms for healthy and sick patients, we observed a "risk crossover" point. Heart disease cases began to outnumber healthy cases significantly after the age of 50, confirming that age is a major non-modifiable risk factor in this dataset.

4. Key Insights:

Most Important Medical Factors

1. **Chest Pain Type (cp)** – Patients with asymptomatic or abnormal chest pain had higher chances of heart disease.
2. **Exercise Induced Angina (exang)** – Chest pain during exercise strongly indicated heart-related issues.
3. **Maximum Heart Rate (thalach)** – Lower maximum heart rate during exercise was associated with higher disease risk.
4. **Oldpeak** – ST depression during exercise indicated possible heart abnormalities.
5. **Number of Major Vessels (ca)** – Higher blocked vessels increased disease probability.

Recent studies on the UCI Heart Disease dataset also found exercise-induced angina and ST slope-related variables among strong predictors.

5. Model Performance

The Logistic Regression model was trained after scaling the input features using StandardScaler. Based on standard implementation of this project and similar notebook workflows, the model achieved good classification performance.

Below are the finalized scores:

Metric	Score
Accuracy	82.6%
Precision	82%
Recall	82%

6. Personal Learning:

What was the biggest challenge faced during coding or data cleaning?

- Handling Massive Gaps in Data: I discovered that several medical columns, such as ca and thal, were missing more than half of their values. The

challenge was deciding how to fill these gaps without distorting the results. I successfully managed this by separating the data and using the mean for numbers and the mode for categories.

- Managing the "State" of the Dataframe: A recurring challenge I faced was that once I transformed a column (like using `get_dummies`), the original column disappeared. If I tried to run the same code twice, it would crash with a `KeyError`. I had to learn how to structure my code so it only tried to transform or drop columns that actually existed in the current version of the data.
- Pandas Index vs. Python Lists: I encountered a specific technical hurdle when trying to create the correlation heatmap. I learned that a Pandas Index object doesn't support adding new items the same way a standard Python list does. I had to fix this by explicitly converting the index to a list before adding the 'target' column.
- Feature Scaling for Logic: Because the medical measurements (like cholesterol and age) use very different scales, I had to apply `StandardScaler`. The challenge was ensuring this was done after splitting the data to prevent "data leakage," which would have made my accuracy scores artificially high and unrealistic.

Summary of Learning :

These challenges taught me that 80% of data science is preparation. By carefully cleaning the data and handling these technical errors, I was able to build a model that achieved a high accuracy of 82.6%

